PHASE 1 MODEL · PHASE 2 FIRMWARE

Edge-AI Worker Safety Monitor

An ESP32 wearable that detects falls, collapses and abnormal vitals on the device, and makes sure the supervisor hears about it.

[Your name] · [Guide / course] · November 2026

THE PROBLEM

What a lone-worker safety device must do

A worker on a site, in a warehouse or on a night shift can fall, faint or overheat with nobody nearby. The device has to notice by itself and get help.

01 Decide on the device

The ESP32 runs the model itself, so detection keeps working when Wi-Fi or mobile coverage is poor.

02 Raise the alarm fast

Alarm about a second after impact, then a 10 s window for the worker to cancel a false one.

03 Stay quiet otherwise

Jogging, stairs and sitting down hard must not trigger it, or workers switch it off.

04 Make sure help is coming

Alerts are retried until the supervisor confirms them, and the worker is told that help is on the way.

HOW IT WORKS

The system at a glance

SENSE

Two sensors

MPU6050 motion sensor at 50 Hz. MAX30102 pulse oximeter for heart rate and SpO2.

→

DECIDE

One ESP32

An int8 neural network for falls; rules for man-down, vitals and sensor faults.

→

ALERT

Body, then Wi-Fi

Buzzer and vibration first. Then an HTTP alert, queued and retried until confirmed.

→

SUPERVISOR

One web page

Shows each alert with posture, impact and vitals. One click acknowledges it.

Everything before the Wi-Fi link runs on the worker's ESP32: **no phone and no cloud** in the decision.

PHASE 1 · MODEL

From motion data to an 18.6 kB model

1 · Data

KFall

32 people, 5,057 recordings of falls and daily activities, sensor at the lower back.

2 · Filter

50 Hz

Causal 5 Hz low-pass, the same filter the chip runs.

3 · Window

2.56 s

128 samples per window, a new window every 0.5 s.

4 · Model

7,169

parameters in a small 1D-CNN; early stopping on validation loss.

5 · Deploy

18.6 kB

int8 TensorFlow Lite; alarm after 2 positive windows in a row.

Split by person: 22 to train, 4 to validate, 6 to test. The test people never appear in training, so the results show how it works on someone new.

PHASE 1 · RESULTS

Results on six people the model never saw

441/441

falls detected

Every fall by the 6 test people, at recording level.

1 of 516

activities raised an alarm

Walking, jogging, stairs, sitting and more: 99.81 % specificity.

0.81 s

median impact to alarm

85 % of falls raise the alarm within 1 s of impact.

The int8 model on the chip gives exactly the same decisions as float32. Five-fold cross-validation by person: 100 % of falls, 99.86 ± 0.28 % specificity.

PHASE 1 · FIXES

Two fixes to the training pipeline

Fix 1 · Stop on validation loss

Validation AUC was already 1.0 after epoch 1, so early stopping kept the epoch-1 weights.

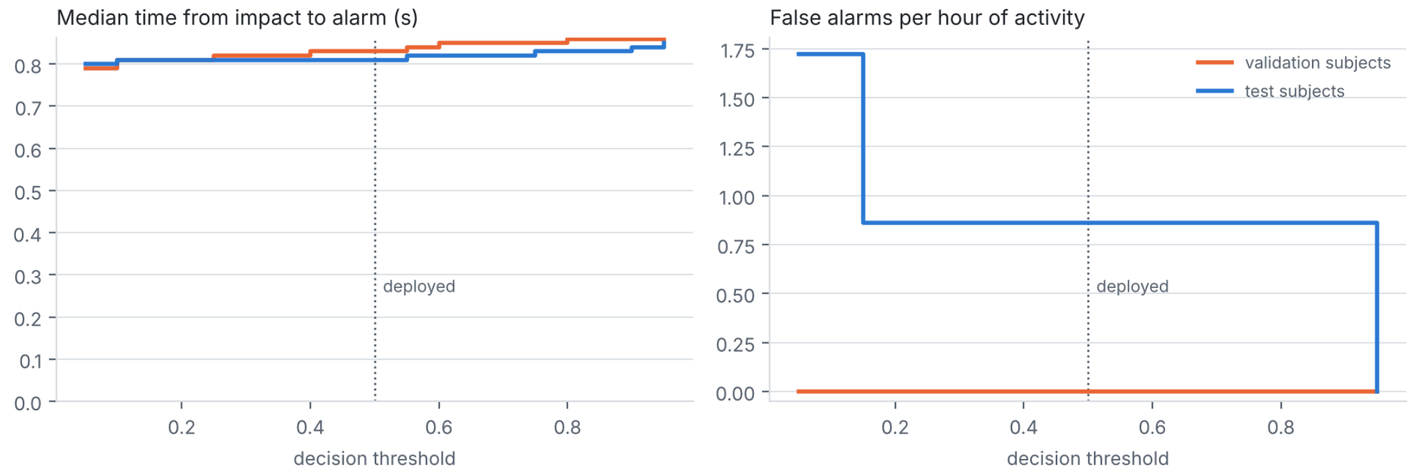
Fix 2 · Middle of the tied range

Every threshold from 0.05 to 0.95 tied at F1 = 1.0. The pipeline now takes the middle, 0.50.

Test people	Original (0.21)	Fix 1 only (0.05)	Both fixes (0.50)
Falls detected	441/441	441/441	**441/441**
False alarms (516 activities)	5	2	**1**
Per-recording accuracy	99.48 %	99.79 %	**99.90 %**
Specificity, 5-fold CV	97.19 ± 3.01 %	99.76 ± 0.34 %	**99.86 ± 0.28 %**

PHASE 1 · DEPLOYMENT CHECK

Every recording replayed the way the chip runs it



Left: median time from impact to alarm. Right: false alarms per hour of activity. Dotted line: the deployed threshold, 0.50.

100 % of falls

at every threshold from 0.05 to 0.95. The threshold only trades false alarms for speed.

0.9 per hour

false alarms: 1 in 1.16 h of hard activity clips. Each first opens the 10 s cancel window.

0.81 s to alarm

median from impact; 95th percentile 1.08 s.

PHASE 2 · FIRMWARE

Firmware on the ESP32

Signal core

Filter, windows, vitals and alert logic in portable C.

Fall model

TensorFlow Lite Micro, int8; 16 KB arena, 7.3 KB used.

Motion sensor

MPU6050 driver, accelerometer and gyro at 50 Hz.

Pulse oximeter

MAX30102 driver for heart rate and SpO2.

Pipeline

Joins sensors, model and rules; watches for sensor faults.

Network

Wi-Fi alerts with a retry queue, acknowledgements, telemetry.

Worker feedback

Buzzer, vibration and the I'm OK button.

Replay

Plays recorded KFall data in place of the sensors.

Fits with room to spare: image 1.01 MB (64 % of the app partition), static RAM 118 KB of 177 KB. Builds for ESP32 and ESP32-S3 with ESP-IDF v5.5.6.

PHASE 2 · SAFETY FEATURES

Six features beyond fall detection

Man-down

Lying still for 30 s without an impact, as after fainting, raises an alert after a pre-alarm.

Motion-aware vitals

Heart-rate alarms pause while the worker moves, so a jogging reading of 165 bpm is ignored.

Sensor fault

If the motion sensor fails, the supervisor is told that fall detection is down.

Skin contact

Vitals are judged only while the oximeter touches skin, never from an empty sensor.

Store-and-forward

Alerts are queued and retried until confirmed. A Wi-Fi dropout delays an alert, never loses it.

Acknowledgement

One click on the supervisor page, and the device beeps twice: help is on the way.

VERIFICATION

Six levels of testing without a board

Level	What is checked	Result
1 · Python model	6 unseen people, plus 5-fold CV by person	441/441 falls, 1/516 false alarms
2 · C port on a laptop	C core against the Python reference	Same decisions, 957 recordings
3 · Model on the chip	int8 self-test inside the ESP32 emulator	28/28 outputs match Python
4 · Firmware build	ESP32 and ESP32-S3, ESP-IDF v5.5.6	Both build; 64 % of flash slot
5 · End to end	250 s scenario in QEMU, supervisor 20 s late	11/11 events, 8/8 messages, 4/4 acks
6 · Supervisor and demo	7 server tests; the demo run in real time	All pass

Every push to GitHub runs all six levels as 7 CI jobs. The latest commit is green.

VERIFICATION · END TO END

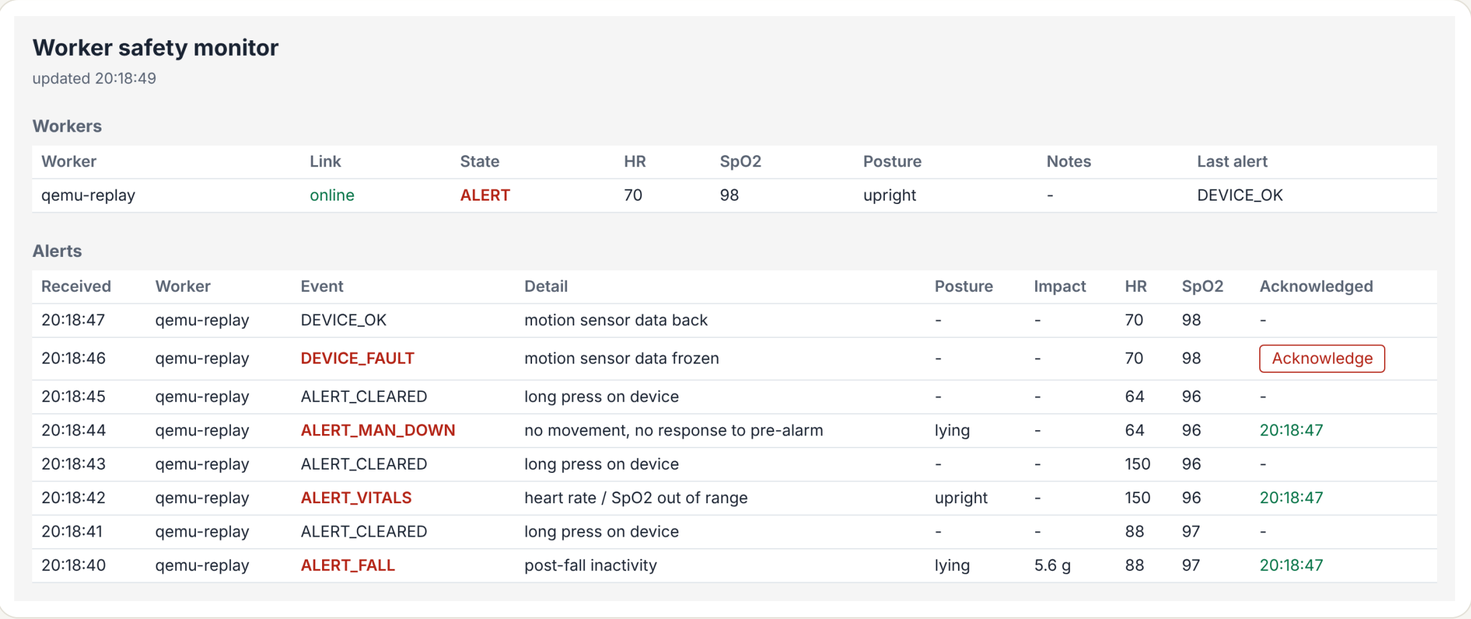
A 250 s scenario, event by event

Start	Situation	What the device does
0 s	Normal work	Nothing (correct)
13 s	Jogging; pulse sensor reads a fake 165 bpm	Nothing: vitals paused while moving
31 s	Fall, worker motionless	FALL_SUSPECTED 39.6 s → **ALERT_FALL** 45.6 s
54 s	Fall, worker presses I'm OK	FALL_SUSPECTED 63.6 s → CANCELLED 66.1 s
77 s	Racing heart at rest	**ALERT_VITALS** 91.1 s
105 s	Slow collapse without impact, lying still	MAN_DOWN_SUSPECTED 138.1 s → **ALERT_MAN_DOWN** 153.1 s
166 s	Lying still, then moves	MAN_DOWN_SUSPECTED 193.1 s → MOVED 200.6 s
204 s	Standing still, upright	Nothing (correct)
244 s	Motion sensor freezes, then recovers	**SENSOR_FAULT** 246.0 s → SENSOR_OK 247.0 s

In the emulator, the firmware raises all 11 expected events at the expected times.

LIVE DEMO

The real firmware, live in the emulator



Worker safety monitor

updated 20:18:49

Workers

Worker	Link	State	HR	SpO2	Posture	Notes	Last alert
qemu-replay	online	ALERT	70	98	upright	-	DEVICE_OK

Alerts

Received	Worker	Event	Detail	Posture	Impact	HR	SpO2	Acknowledged
20:18:47	qemu-replay	DEVICE_OK	motion sensor data back	-	-	70	98	-
20:18:46	qemu-replay	DEVICE_FAULT	motion sensor data frozen	-	-	70	98	Acknowledge
20:18:45	qemu-replay	ALERT_CLEARED	long press on device	-	-	64	96	-
20:18:44	qemu-replay	ALERT_MAN_DOWN	no movement, no response to pre-alarm	lying	-	64	96	20:18:47
20:18:43	qemu-replay	ALERT_CLEARED	long press on device	-	-	150	96	-
20:18:42	qemu-replay	ALERT_VITALS	heart rate / SpO2 out of range	upright	-	150	96	20:18:47
20:18:41	qemu-replay	ALERT_CLEARED	long press on device	-	-	88	97	-
20:18:40	qemu-replay	ALERT_FALL	post-fall inactivity	lying	5.6 g	88	97	20:18:47

The supervisor page (supervisor.py) during an emulator run.

1 Start the device

```
docker start -ai safety-demo
```

2 Open the supervisor

localhost:8080 in a browser, next to the device log.

3 Acknowledge

Click on each alert. The device beeps twice: help is on the way.

NEXT STEPS

From emulator to a worn prototype

Hardware

- Build it: ESP32, MPU6050, MAX30102, buzzer, vibration motor and a button, about USD 15–20
- Measure inference time and battery life on the real chip
- Match the sensor's axes to KFall's at the lower back
- Keep queued alerts in flash so a reboot cannot lose them

Validation

- Controlled falls onto a mattress, with someone present
- Full-shift wear tests to count real false alarms per hour
- Compare heart rate and SpO2 with a reference oximeter
- Retrain with recordings from the device itself if they differ from KFall

SUMMARY

Phase 1 · Model

An 18.6 kB int8 model: 441/441 falls of unseen people, 1 false alarm in 516 activities.

Phase 2 · Firmware

ESP32 firmware with man-down, motion-aware vitals, fault checks and alerts that are never lost.

Verified

Six levels of tests on every push, and a live demo in the emulator. Next: a real board.

Questions?

github.com/chaithr45/Snehitha_Proj · branch claude/great-euler-gnc1oj